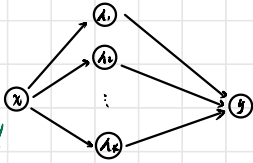


Shallow neural network

→ when we have only one layer, we normally say it's shallow



Suppose input is 1d

$$\text{where } h_i = \theta_{i0} + \theta_{i1}x \text{ for } i = 1, \dots, k$$

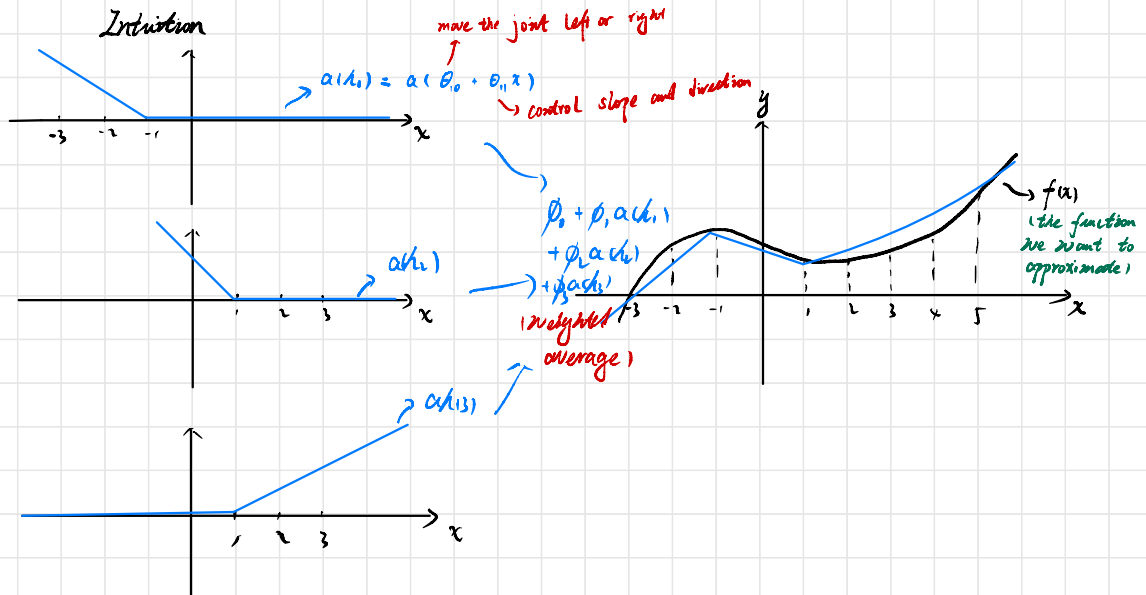
$$y = \phi_0 + \phi_i \alpha(h_i)$$

→ activation function.
below we consider it
as ReLU if not otherwise
specified.

1. Universal approximation theorem

It states that for any continuous function, there exists a shallow neural network that can approximate this function to any specified precision.

Intuition

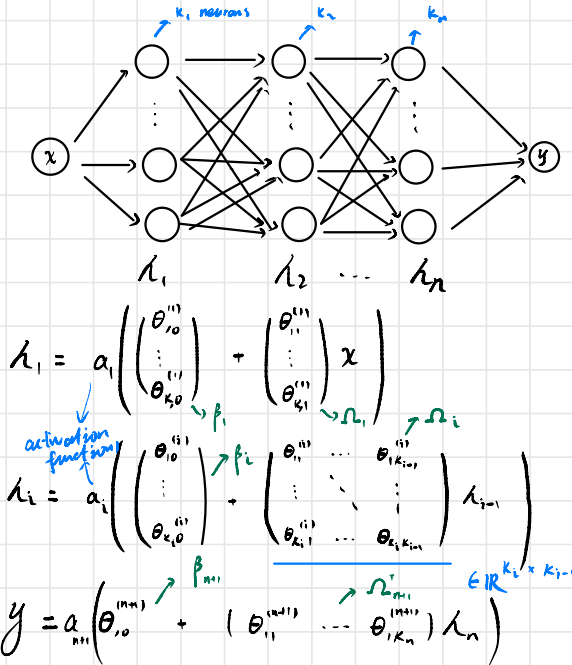


2. Number of possible linear regions

When input is 1d, in theory there are at most $D+1$ linear regions ↗ one additional neuron will add at most one joint

When input $x \in \mathbb{R}^D$, and the hidden units $k=D$, there are at most 2^D linear regions

Deep neural network



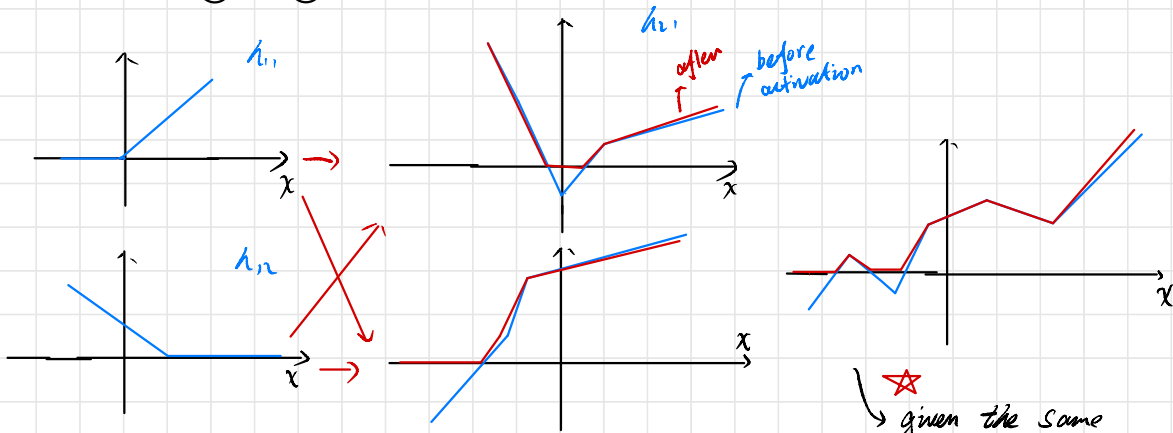
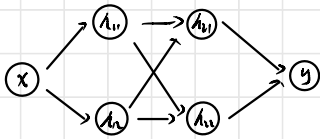
Equivalently, we have

$$y = a_{n+1} \left(\beta_{n+1} + \Omega_{n+1} \cdot a_n \left(\beta_n + \Omega_n \cdot a_{n-1} \left(\beta_{n-1} + \Omega_{n-1} \cdot x \right) \right) \dots \right)$$

Deep neural network can also approximate any continuous functions.

In theory however, deep neural network can create much more possible
(exponentially more)
regions given fixed number of parameters

Consider a simple example



☆ given the same number of parameters, deep neural network creates many more linear regions